

Transcale: Embodiment Transition Toward Multi-verse Exploration

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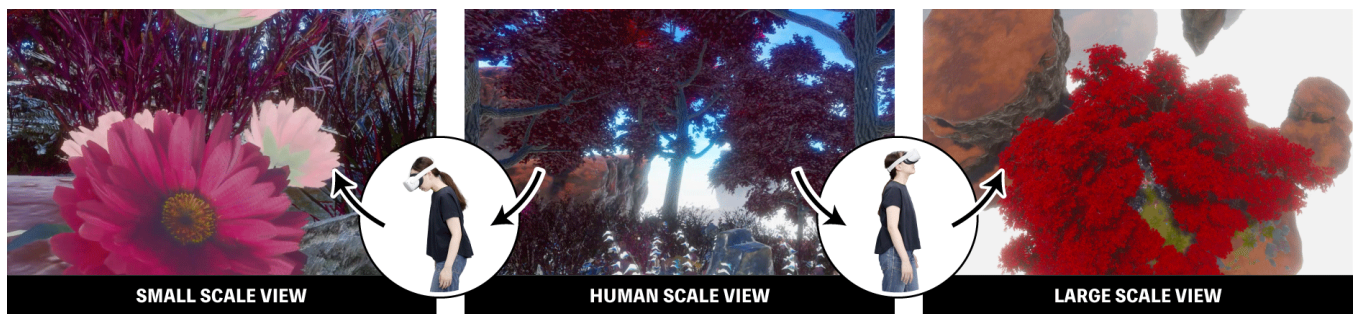


Figure 1: Transcale is a VR experience that allows users to manipulate their body scale by linking their movements to scale changes. In this experience, when a user walks while looking down, the scale becomes smaller, and when they walk while looking up, the scale becomes larger.

ABSTRACT

Transcale is a VR experience where the users can change the body scale smoothly while walking. By continuously modifying their perceived body scale, this interaction method provides the users to explore the virtual world from various points of view from a dwarf to a giant.

CCS CONCEPTS

• **Human-centered computing** → *Systems and tools for interaction design.*

KEYWORDS

embodiment, virtual reality, interaction design, multi scale, body ownership

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1 INTRODUCTION

Our perception of scale is based on our understanding of the physical reality. A squirrel is a small creature, a mountain is a large thing, etc. We judge the scale of things according to our previous knowledge and experience. Therefore, it is difficult for us to accurately imagine the world on a scale far removed from our daily experience, for example, the world as seen by creatures such as plankton or whales, which are significantly different in size from humans.

Conversely, in a virtual environment, it's possible to transcend these physical limitations and represent worlds of diverse scales. From micro-worlds, such as virus size, to macro-worlds that exceed the size of the Earth, similar to Power of Ten, a classic educational video showing this vast range of scales¹.

Recently, VR technologies have been applied to experience and learn about worlds on scales that aren't usually perceptible [Wu et al. 2022]. Research on scale changes in VR has been conducted for various purposes, including studies that transition between robots of different sizes [Izumihara et al. 2019], and studies on transitions for moving to viewpoints of different scales within VR [Lee et al. 2022].

Thus, there have been many studies and productions of contents based on perspectives of different scales from daily experience.

¹Powers of Ten (1977), <https://www.youtube.com/watch?v=0fKBhvDjuy0>

However, a method for freely transitioning between viewpoints at different scales without losing the physical sensation of being there has not been clarified.

Therefore, in this study, we created a transition design in which the user's own body size changes in conjunction with the user's body movements (Figures 1). We thought that such a design would enable transitions that do not detract from the user's physical experience. We define the experience of using this method as Transcale.

2 PROPOSED METHOD

2.1 TRANS-SCALE EMBODIMENT

To recognize one's virtual body as their own in VR space, it's important to actively move that body.

In Trans-scale Embodiment, users can manipulate their own body scale by linking their body movements to scale changes. For instance, in this experience, we employed methods that linked movements, like head movements and walking, to scale changes.

2.2 INTERACTION DESIGN

We developed a system where the user wears a head-mounted display (HMD). When users walk while looking down, their virtual body shrinks, and when they walk while looking up, it expands (Figures 2). The body size adjusts gradually in relation to the walking distance, enabling seamless transitions between viewpoints of different scales. This continuous adjustment provides users the sensation of altering their own body scale. Additionally, since both the walk distance and the interpupillary distance (IPD) change in proportion to the spatial scale, users are likely to perceive a modification in their body size.

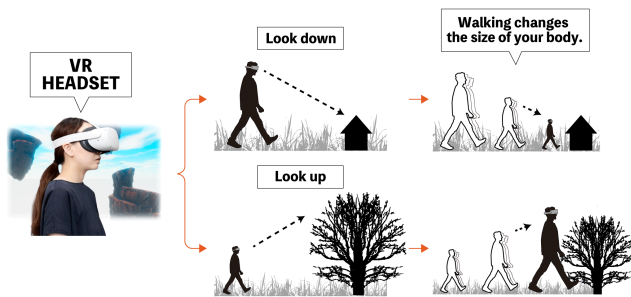


Figure 2: Interaction design for scale transition

3 EXPERIENCE DESIGN

In the Transcale experience designed for this project that is an immersive experience that allows users to walk and explore within a VR environment (Figures 3).

Various objects such as plants, birds, and butterflies are placed within the VR environment. Moreover, to stimulate the user's curiosity and encourage exploration, the world of the piece is made fantastical. In the world of this project, users can explore as they wish not only from a human-sized perspective, but also from any scale they prefer. In addition, several objects were placed that are difficult to find without changing the body scale. For example, a

miniature house is positioned in a spot that's hard to see at human size, but can be discovered by shrinking oneself.

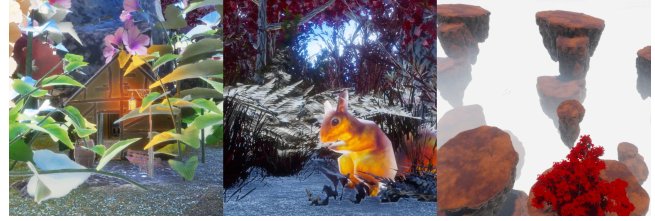


Figure 3: In scenes in Transcale experience, objects of different scales, such as a squirrel, a miniature house, and a floating island, are placed.

4 FUTURE WORKS

Research is also being conducted to create illusions of scale changes in real spaces using tools such as panoramic cameras and drones [Hashizume et al. 2018]. The proposal in this paper focuses on VR environments. However, by integrating panoramic cameras with moving objects, we think experiencing scale changes in real spaces, not just in VR, becomes possible.

Furthermore, if we apply the technique of Trans-scale Embodiment, which combines body movements with visual changes, we believe it's possible to transition into worlds that involve physical transformations, such as a two-dimensional world or a world with different gravity, not just changes in scale.

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